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Professor Rashmi

CS6065 Introduction to Cloud Computing

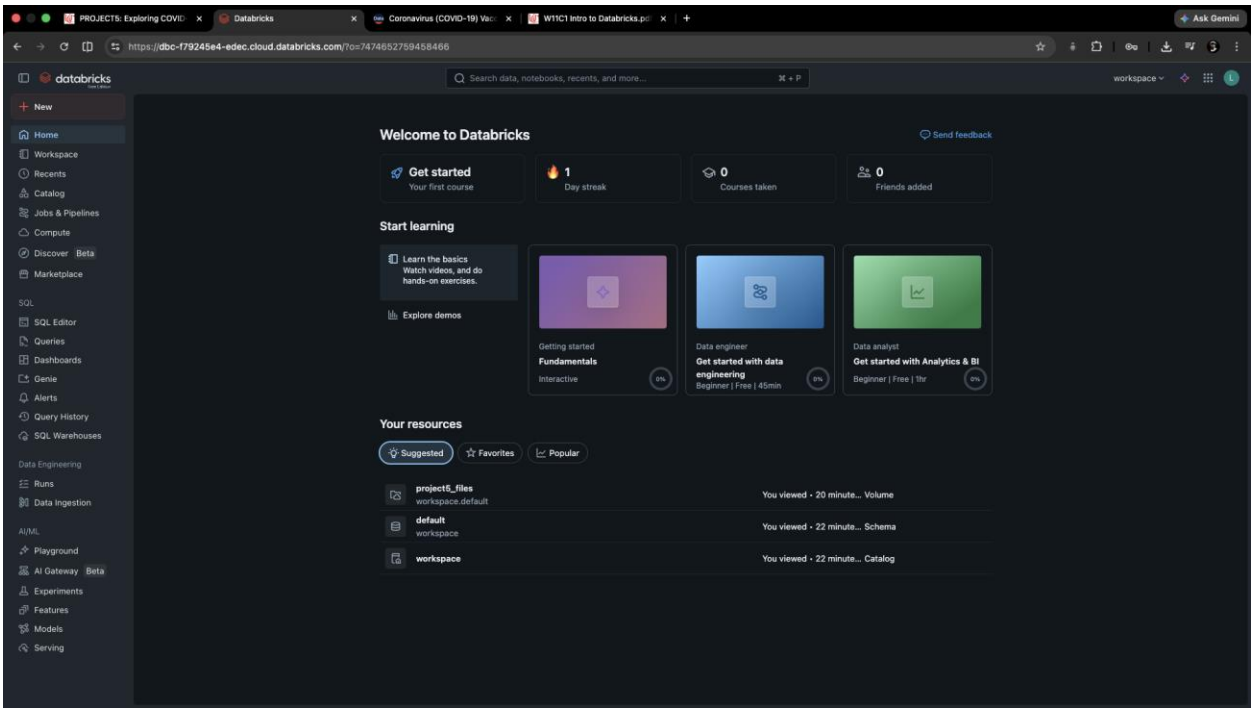
Assignment 5 (Exploring COVID-19 Data using Databricks)

March 26, 2026

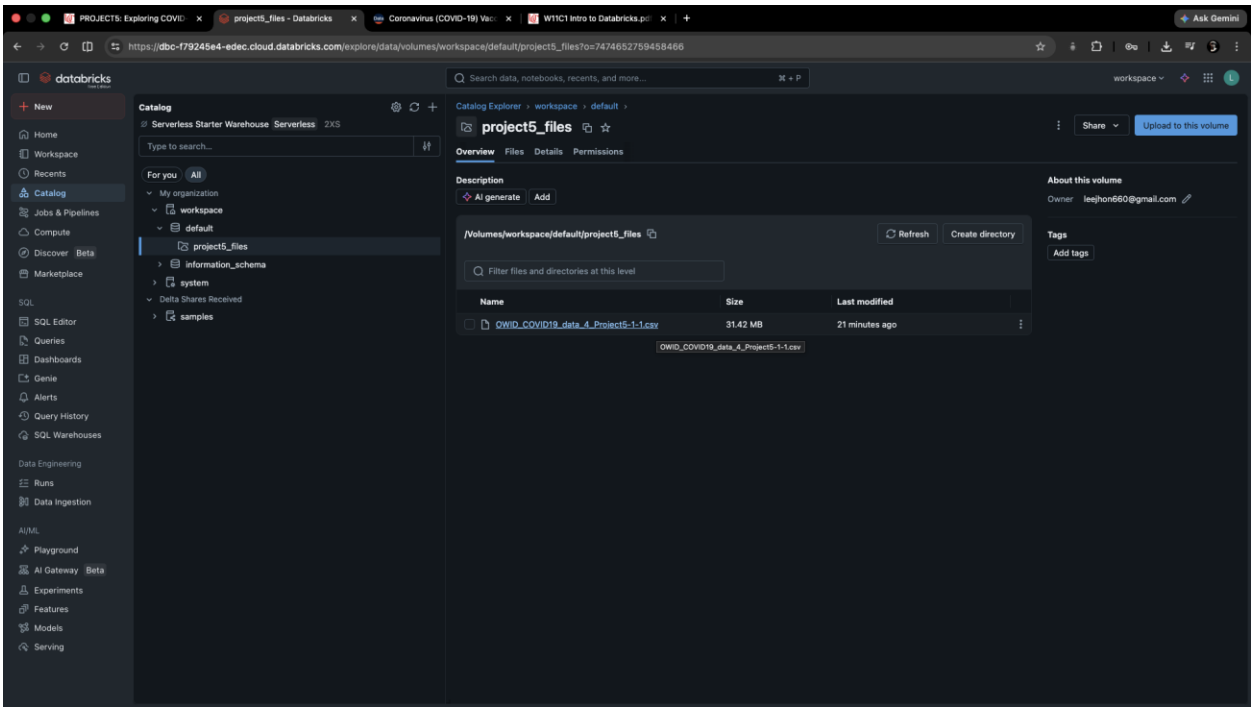
Code on Github: https://github.com/MansurPro/covid-vaccination-impact-analysis

1. Install Databricks and upload the data

1.1 Databricks workspace

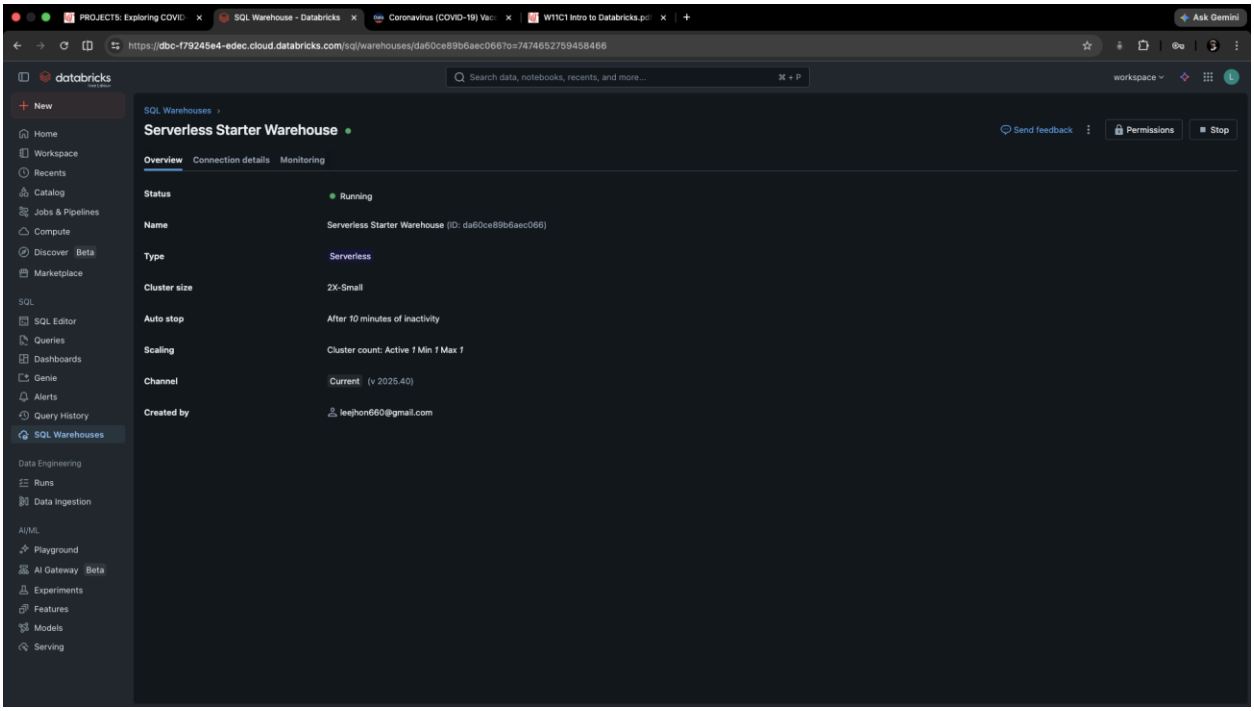


1.2 Upload the CSV file (Dataset)

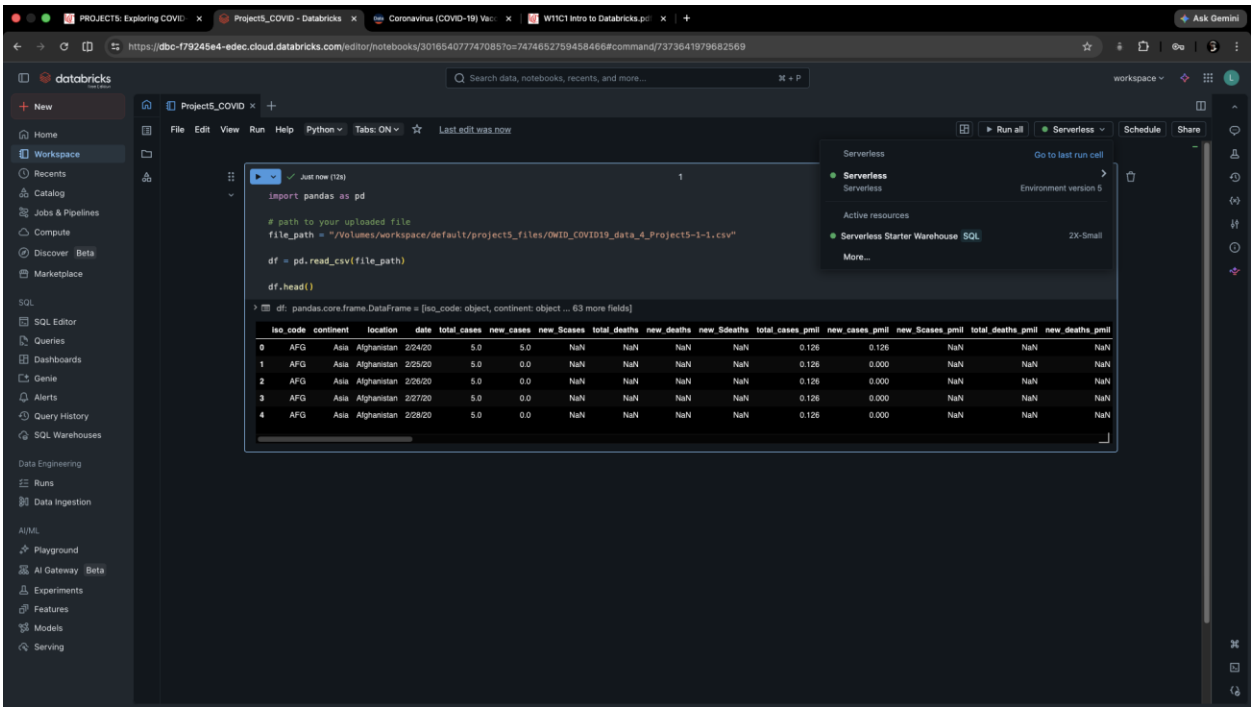


2. Load the data file using Pandas as needed

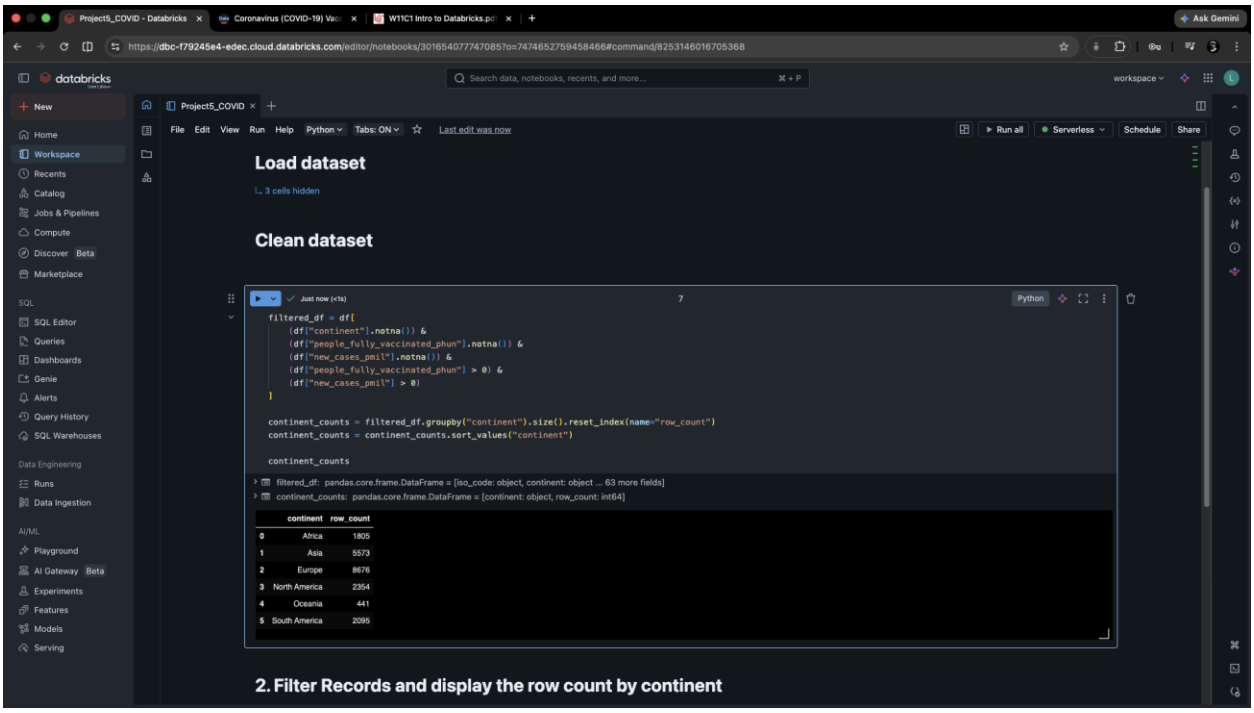
2.1 Warehouse running



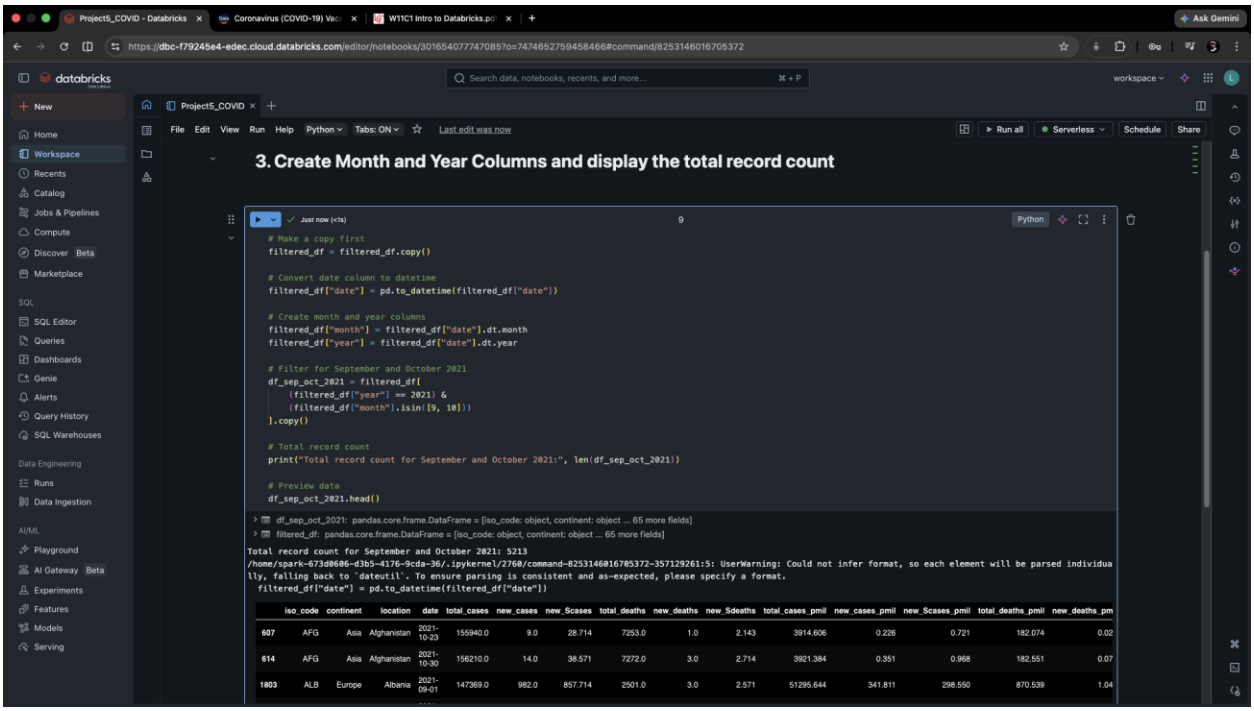
2.2 Notebook showing dataset loaded



3. Filter Records and display the row count by continent

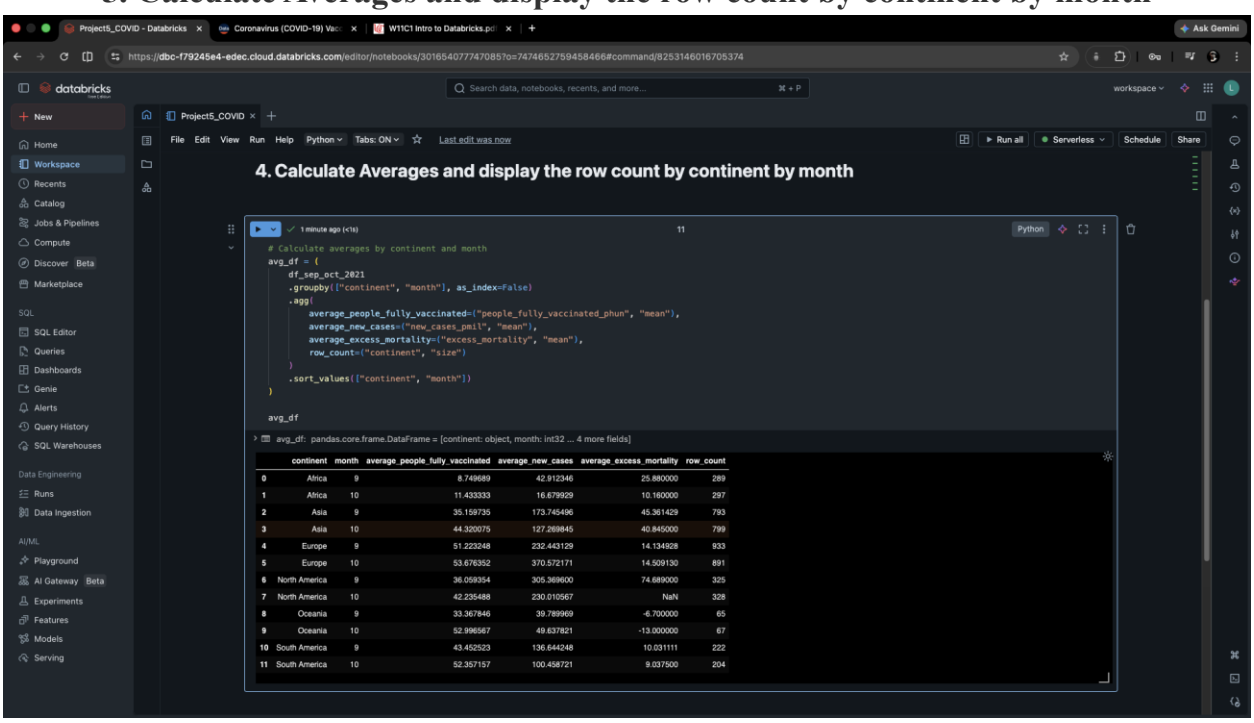


4. Create Month and Year Columns and display the total record count



	iso_code	continent	location	date	total_cases	new_cases	new_5cases	total_deaths	new_deaths	new_5deaths	total_cases_pmil	new_cases_pmil	new_5cases_pmil	total_deaths_pmil	new_deaths_pm
607	AFG	Asia	Afghanistan	2021-10-23	155940.0	9.0	28.714	7253.0	1.0	2.143	3914.606	0.226	0.721	182.074	0.02
614	AFG	Asia	Afghanistan	2021-10-30	156210.0	14.0	38.571	7272.0	3.0	2.714	3921.384	0.351	0.968	182.551	0.07
1803	ALB	Europe	Albania	2021-09-01	147369.0	982.0	857.714	2501.0	3.0	2.571	51295.644	341.811	298.550	870.539	1.04

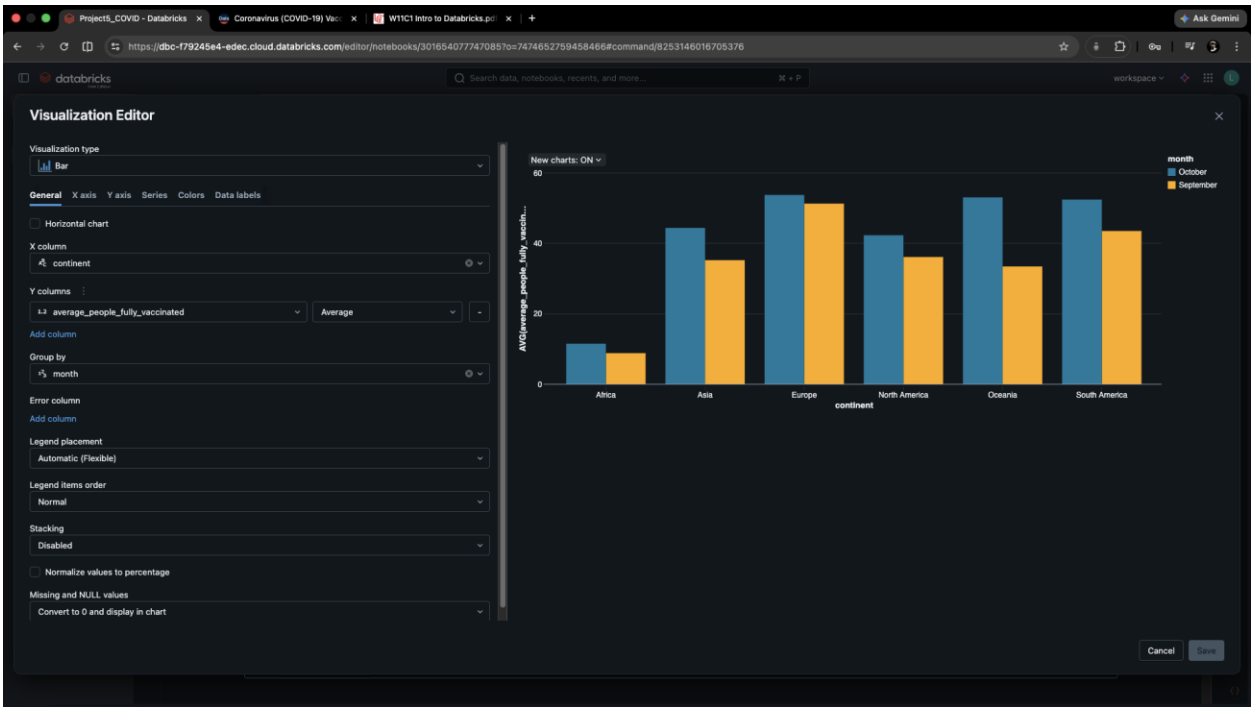
5. Calculate Averages and display the row count by continent by month



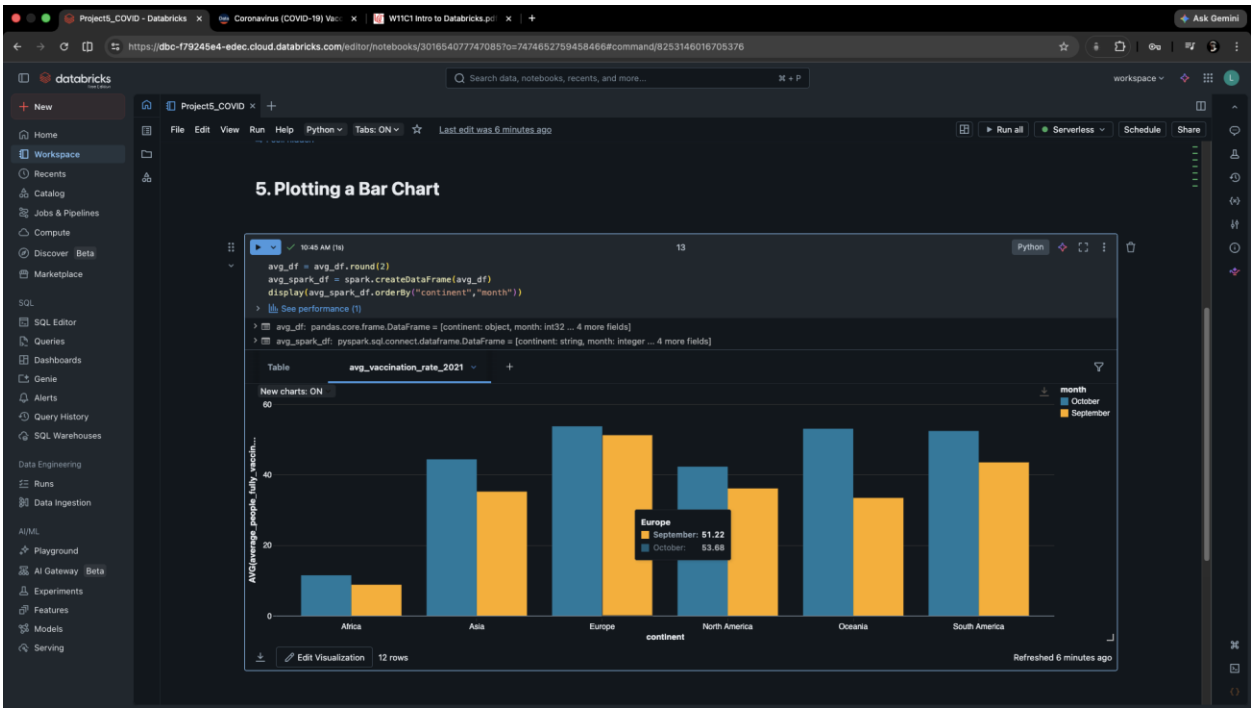
	continent	month	average_people_fully_vaccinated	average_new_cases	average_excess_mortality	row_count
0	Africa	9	8.749689	42.912346	25.880000	289
1	Africa	10	11.433333	16.679929	10.160000	297
2	Asia	9	35.159735	173.745496	45.361429	793
3	Asia	10	44.320075	127.269845	40.845000	799
4	Europe	9	51.223248	232.443129	14.134928	933
5	Europe	10	53.676352	370.572171	14.509130	891
6	North America	9	36.059354	305.369600	74.689000	325
7	North America	10	42.235488	230.010567	NaN	328
8	Oceania	9	33.367846	39.789969	-6.700000	65
9	Oceania	10	52.996567	49.637821	-13.000000	67
10	South America	9	43.452523	136.644248	10.031111	222
11	South America	10	52.357157	100.458721	9.037500	204

6. Plotting a Bar Chart.

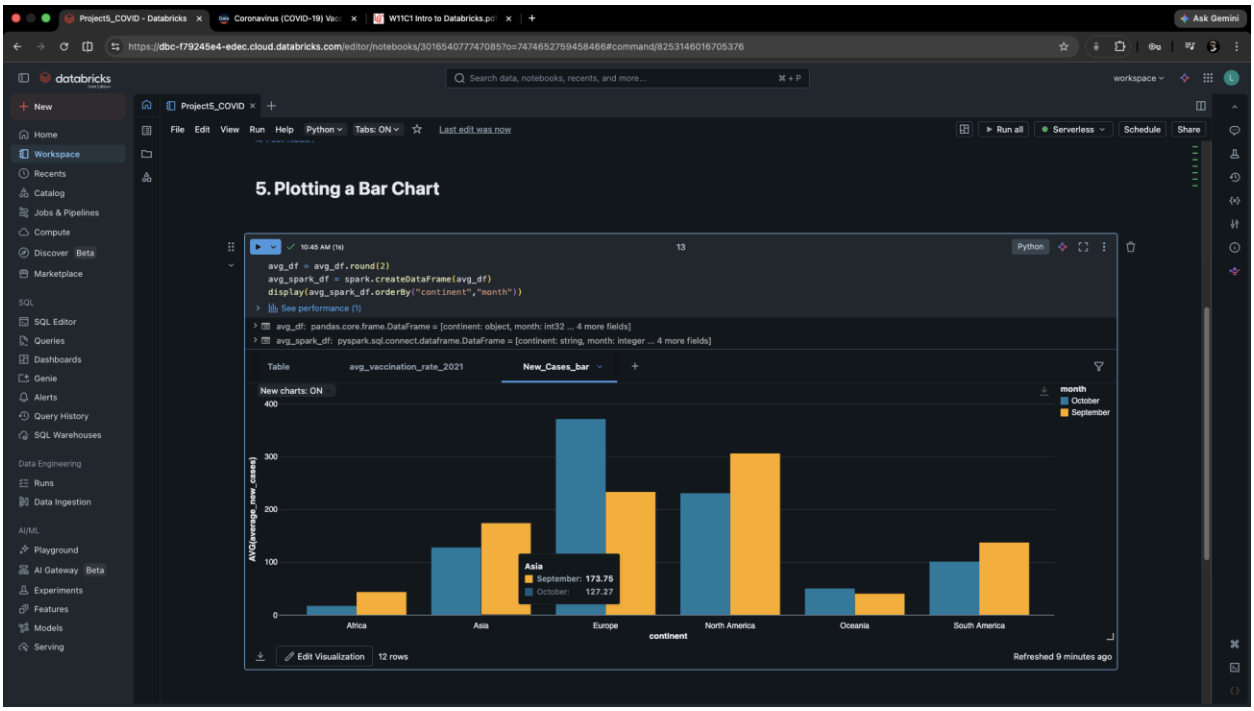
6.1 Visualization Editor settings



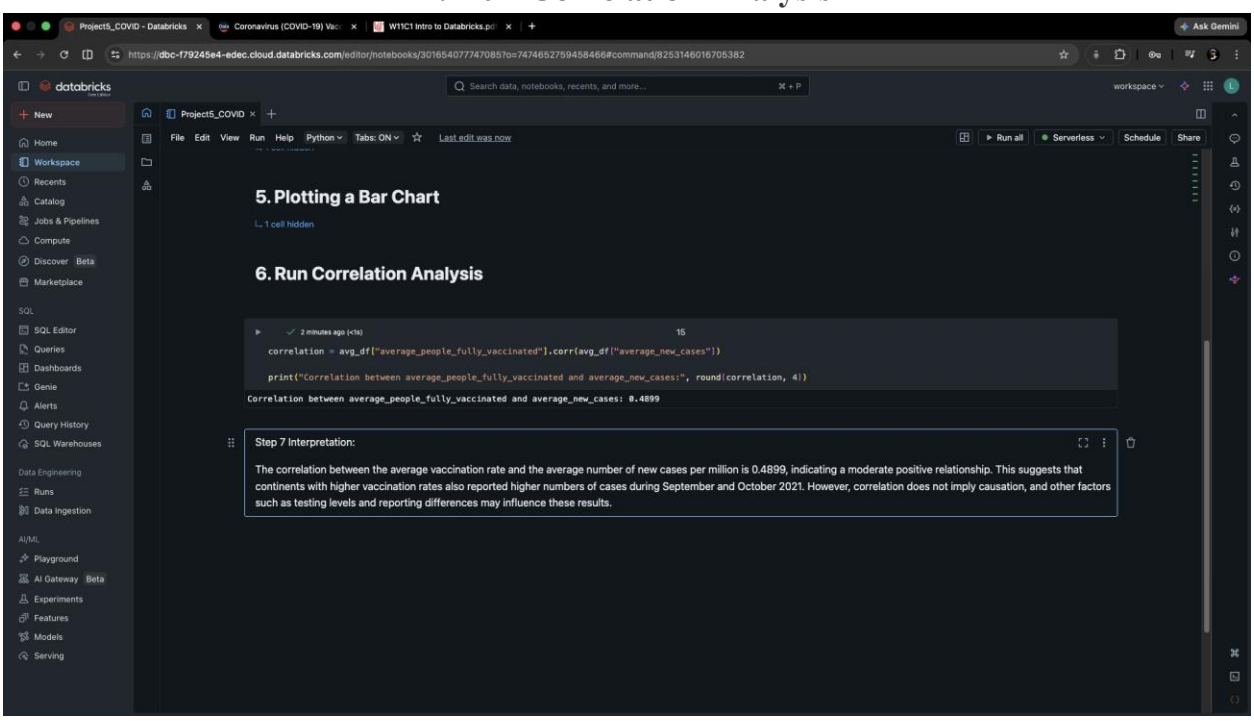
6.2 Vaccination bar chart



6.3 New Cases bar chart



7. Run Correlation Analysis

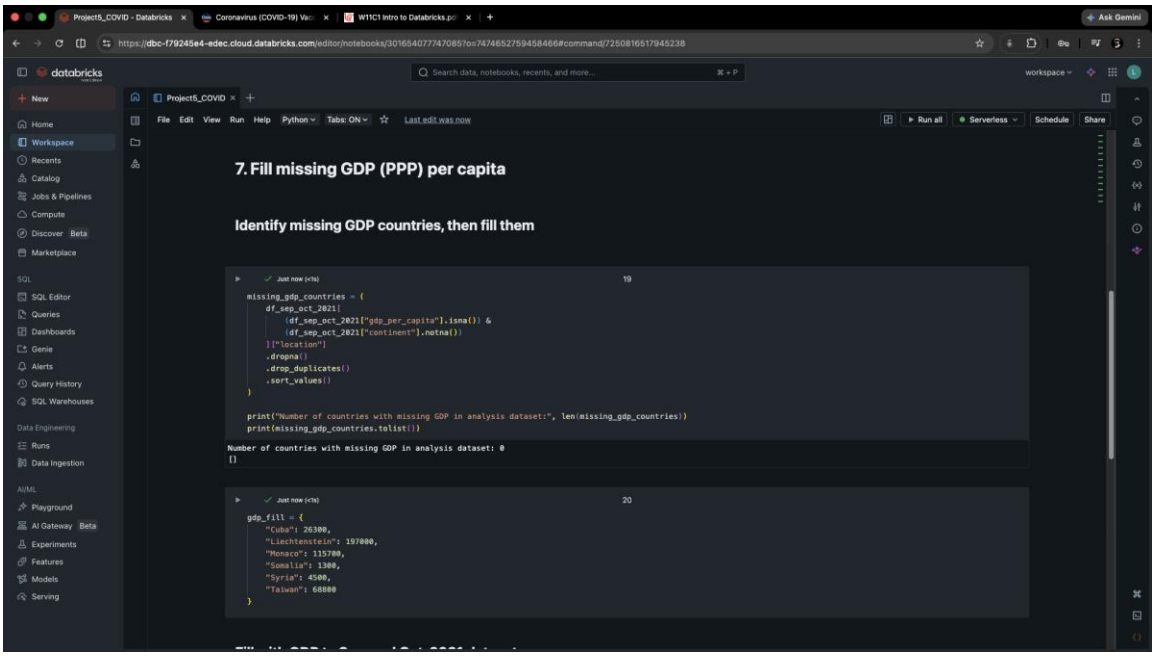


Step 7 Interpretation:

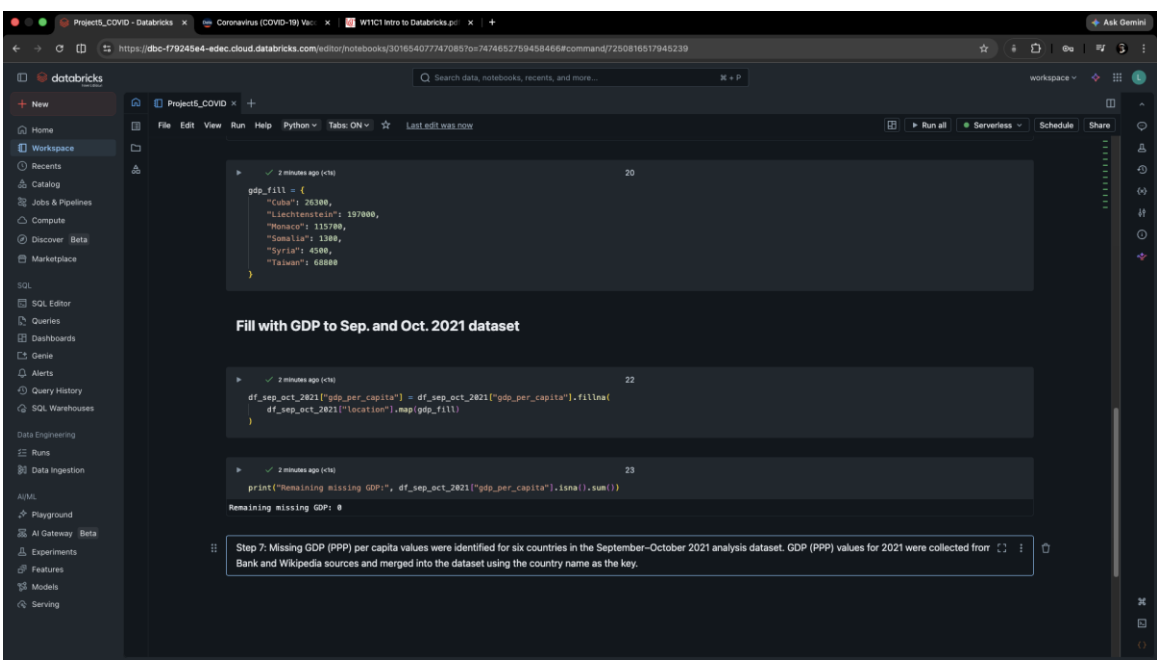
The correlation between the average vaccination rate and the average number of new cases per million is 0.4899, indicating a moderate positive relationship. This suggests that continents with higher vaccination rates also reported higher numbers of cases during September and October 2021. However, correlation does not imply causation, and other factors such as testing levels and reporting differences may influence these results.

8. Fill missing GDP (PPP) per capita

8.1 List of missing GDP countries

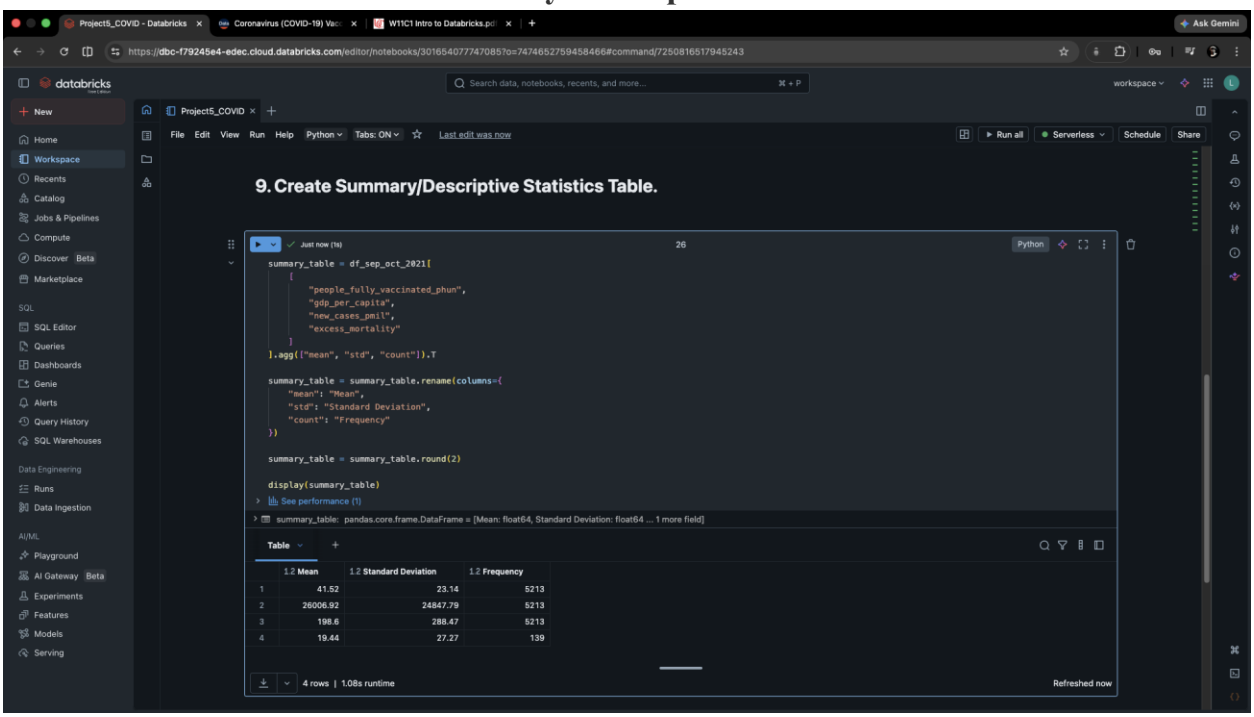


8.2 Code filling the values



Step 8: Missing GDP (PPP) per capita values were identified for six countries in the September–October 2021 analysis dataset. GDP (PPP) values for 2021 were collected from World Bank and Wikipedia sources and merged into the dataset using the country name as the key.

9. Create Summary/Descriptive Statistics Table.



10. Reporting Results.

The results of this analysis show that higher COVID-19 vaccination rates were not associated with lower reported case numbers during September and October 2021. The correlation between average vaccination rate and average new cases per million was 0.4899, which indicates a moderate positive association. The bar charts support this result by showing that continents with higher vaccination levels often still had relatively high case counts. This does not imply that vaccination increased infections, but rather that other factors, such as stronger testing systems, better reporting, higher population movement, and different outbreak timing, may have affected the results. Vaccination may still have helped reduce severe outcomes and excess mortality, but that relationship would need additional correlation or regression analysis with the excess mortality variable. Therefore, this project suggests that vaccination rates alone were not enough to explain lower case counts, and the relationship with mortality should be interpreted with caution.